Philipp Benjamin Aretz

Center for Theoretical Physics, Massachusetts Institute of Technology · Cambridge, MA, USA

Research interests

Effective field theory and precision QCD; factorization and resummation for collider observables; medium effects in heavy-ion collisions; mathematical structure of quantum field theory.

Education

PhD in Physics, Massachusetts Institute of Technology, Cambridge, MA, USA 2024–present

MASt in Mathematics (Part III), University of Cambridge, Cambridge, UK 2023–2024
Pass with Distinction, 92% — ranked 9th of 294

BSc in Physics, Heidelberg University, Heidelberg, Germany 2020–2023
1.0 (best possible)
Thesis: *The Keldysh Functional Integral and Bounds on Truncated Expectation Values* (advisor: Prof. Manfred Salmhofer)

Abitur, Goethe-Schule Bochum, Bochum, Germany 2020
1.0 (best possible)

Research positions

Research Assistant, Heidelberg University, Heidelberg, Germany 2022–2023
CRC/TRR 191 Symplectic Structures in Geometry, Algebra and Dynamics
Differential delay equations: periodic delay orbits and smooth orbit families. Magnetic billiards: KAM-based extension near convex boundaries.

Tutor, Heidelberg University, Heidelberg, Germany 2021–2022

Financial Consulting Intern, Kartak & Partner, Germany 2021

Publications and preprints

Published

P. B. Aretz, M. Salmhofer, *A rigorous Keldysh functional integral for fermions*, J.Statist.Phys. 193 (2026) 28, arXiv:2508.01787 [math-ph].

P. Albers, **P. Aretz**, I. Seifert, *Families of periodic delay orbits*, J. Differential Equations 410 (2024) 228-250, arXiv:2304.07550 [math].

Preprints

P. B. Aretz, K. Lee, S. T. Schindler, I. W. Stewart, *Factorization of elastic, single, and double diffractive pp scattering*, arXiv:2607.07788 [hep-ph].

Talks

Conference Talks

Diffraction Across Colliders, SCET 2026, Seoul, South Korea 2026

Posters

Diffraction at the LHC, ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna 2026

Generalised Detectors in Heavy Ion Collisions, ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna 2026

Awards and funding

Jennings Prize, University of Cambridge 2024

DAAD Scholarship, Deutscher Akademischer Austauschdienst 2023

Wolfson College Scholarship, Wolfson College, Cambridge 2023

Service and leadership

President MIT Rowing Club, MIT Rowing Club 2026–present

Treasurer, Edgerton House, MIT 2025–2026

Organiser, First Year Seminar, Massachusetts Institute of Technology 2024–2025

Technical skills

Python (NumPy, pandas, TensorFlow); C++ (ROOT, Pythia); Mathematica; MATLAB; LaTeX.

Languages

German (native); English (C2, Cambridge Grade A, TOEFL 117/120); French (basic); Spanish (basic); Latin (Latinum).